

GridWatch 3.0 - Product Specification

Demonstration document - fictional data

Overview

GridWatch 3.0 is Meridian Labs' monitoring suite for utility-scale solar farms. It ingests inverter telemetry at 1-second resolution, detects anomalies with a hybrid rules-plus-ML engine, and issues maintenance work orders automatically.

Functional Requirements

FR-1: The system shall ingest at least 50,000 telemetry points per second per site. FR-2: Anomaly alerts shall be raised within 30 seconds of signal deviation. FR-3: Operators shall be able to acknowledge, assign, and close alerts from the web console. FR-4: Historical data shall be queryable for a minimum of 7 years.

Non-Functional Requirements

Availability target is 99.9 percent monthly. All data is encrypted in transit (TLS 1.3) and at rest (AES-256). The web console must render dashboards in under 2 seconds at the 95th percentile. Role-based access control distinguishes Operator, Engineer, and Admin roles.

Release Plan

Beta begins March 2026 with three pilot sites in Rajasthan. General availability is targeted for July 2026. Migration tooling from GridWatch 2.x will be provided, with dual-running supported for 90 days after upgrade.